

DeribitVerdictEngine — Trader's Guidebook

Who this is for: Experienced BTC perpetual scalpers on Deribit.

What this is not: A programming manual. All implementation detail has been removed.

How to use it: Read top-to-bottom on each engine run. Every section answers:

"What does this tell me right now, and what should I do with it?"

The Output — Top to Bottom

Every time the engine runs, it prints its output in this order:

1. **Verdict Block** — the headline call and confidence level.
 2. **ATR Entry Levels** — volatility context and reference trade frame.
 3. **Kelly Sizing** — directional conviction expressed as a sizing hint.
 4. **Dynamic Norms** — how the engine has adapted its thresholds to current conditions.
 5. **Regime** — the broader market structure context (trending vs ranging).
 6. **Core Signals (1m)** — momentum, mean-reversion, and volume on the execution timeframe.
 7. **VWAP** — where price sits relative to session fair value.
 8. **BBW / TTM Squeeze** — whether energy is coiling or releasing.
 9. **EMA Ribbon** — trend alignment across timeframes.
 10. **Market Structure** — swing highs/lows, Donchian breakouts, and OBV trend.
 11. **Open Interest** — whether new money is entering the market.
 12. **Order Flow** — who is actually buying and selling in real time.
 13. **Liquidations** — forced exits that can fuel or exhaust a move.
 14. **MTF Gate** — the 15-minute timeframe alignment check.
 15. **Funding** — the cost-of-carry signal for perpetuals.
 16. **Signal Breakdown Table** — the full score ledger.
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1. Verdict Block

VERDICT

What it shows: The engine's final directional call — the one-word answer to "long, short, or stand aside?"

What to watch for:

- **STRONG LONG / STRONG SHORT** — high signal density; this is your primary trade trigger.
- **MEDIUM LONG / MEDIUM SHORT** — solid setup, medium confidence; valid for entry per trader-profile rules. *(v55, 2026-07-21: renders MEDIUM x on-screen; **stored** as bare LONG / SHORT on the CSV, bridge payload, and eval cache — the display-only change kills the "STRONG vs LONG" ambiguity on the ladder.)*
- **WEAK LONG / WEAK SHORT** — partial setup only; use as context, not an entry trigger.
- **NO TRADE** — neither direction has enough confirmation; stand aside.
- **NO TRADE [WEAK LONG] / NO TRADE [WEAK SHORT]** — a lean exists but a hard gate killed it (regime mismatch or 15m misalignment); the direction is noted but the trade is blocked.

Example: **STRONG LONG** means the majority of indicators are aligned bullish with high cross-confirmation density — this is the cleanest entry signal the engine produces. **NO TRADE [WEAK SHORT]** means the engine saw a faint short lean but the 15-minute timeframe said no; respect the block and wait.

CONTEXT

What it shows: A warning label that tells you *why* the verdict might be weaker than its tier suggests. Only appears when something is off — if this line is absent, the setup is clean.

What to watch for:

- **MOMENTUM_FADING** — price is still moving in the verdict direction, but the driving force is rolling over. Tighten your profit target and don't chase entries.
- **FLOW_UNCONFIRMED** — the technical structure looks right but real buying/selling pressure hasn't shown up yet. High fakeout risk; wait for flow to confirm.
- **STRUCTURALLY_WEAK** — neither price structure nor order flow has enough evidence. The verdict is a weak lean at best; skip or size tiny.
- **BELOW_MIN_MOVE (v35)** — the verdict had a real lean but the realistic move (after target capping) is too small to be worth trading — under ~0.08% of price, roughly \$50 at \$62k BTC. Displays as **NO TRADE** regardless of how the raw score looked. Most common in low-ATR Asia/London conditions. Not a bug — the engine is refusing to call a trade it can't size meaningfully.
- **No CONTEXT line** — the setup is balanced and confirmed on both axes. This is the cleanest version of the verdict.

Example: **WEAK LONG** with **MOMENTUM_FADING** shown = borderline, lean toward skipping. **WEAK LONG** with no **CONTEXT** line = cleaner setup, cautious entry is more justifiable.

CONFIDENCE

What it shows: A plain-English tier label — **HIGH**, **MEDIUM**, **LOW**, or **N/A** — that maps directly to the verdict strength.

What to watch for:

- **HIGH** or **MEDIUM** — the only tiers where your trader-profile rules say act.
- **LOW** — informational only; treat like a watch-and-wait signal.
- **N/A** — verdict is **NO TRADE**; no tier was cleared, or a hard gate fired.

Example: **CONFIDENCE: MEDIUM** on a **LONG** verdict = valid entry signal. **CONFIDENCE: LOW** on a **WEAK SHORT** = note it but don't trade it.

SCORE

What it shows: The raw point tally for both long and short sides, compared to the maximum possible score for the current market regime.

What to watch for:

- The format is **Long N/M | Short N/M** where M is the regime ceiling (15 in choppy markets, 19–20 in trending/ranging markets).
- When the market is in a choppy (**TRANSITIONAL**) regime, you may see an (**eff.E**) effective score that is lower than the raw score — a penalty has been applied.
- A large gap between raw and effective score means the ADX is borderline; factor that uncertainty into your sizing independently of the verdict tier.
- The side with the higher effective score wins the verdict.

Example: **Long 12/20 (eff.10) | Short 3/20 | TRANSITIONAL penalty: -2** — the long side has a solid raw score, but the choppy regime shaved 2 points off. The long lean is real but you're fighting uncertain conditions; size down.

TIME

What it shows: The exact local time (UTC+8) when this engine run was processed.

What to watch for:

- Use this to confirm the output is fresh, especially if you're running on an auto-refresh schedule.
- Cross-reference with session boundaries: Asian session is 08:00–15:59 local, London is 16:00–20:59 local, New York is 21:00 local onward.
- A stale timestamp means the engine may not have polled fresh data — treat the verdict with caution.

Example: If your auto-run fires every 60 seconds but the timestamp shows 3 minutes ago, something stalled. Refresh manually before acting.

LAST TRANSACTED PRICE

What it shows: The price of the most recent actual trade on the exchange, captured live — separate from the candle close price used in calculations.

What to watch for:

- Compare this to the entry price in the ATR block below it. If they diverge significantly, a fast move happened in the seconds since the last candle closed.
- A large gap (e.g., >\$50 on BTC at current volatility) means an impulse is in progress right now — your entry levels are already stale.
- N/A means the live trade feed returned empty; treat the whole run with caution.

Example: ATR entry shows **76038.0** but Last Transacted Price shows **76120.0** — price has already moved \$82 in the time since the candle closed. If you were planning a long, you're now chasing; reassess.

HOLD / EXIT

What it shows: Real-time guidance for managing an open position. Only appears when you have a position tracked in the engine.

What to watch for:

- **EXIT -- microstructure deterioration** — multiple flow signals just flipped against you simultaneously; you're likely already late to exit.
- **EXIT -- momentum break** — the momentum oscillator just crossed zero against your position; structural exit trigger.
- **TAKE PROFIT -- extreme momentum, tighten stops** — you're running an outlier extension; don't get greedy, bring your stop up.
- **EVALUATE -- ...** — a single warning signal; check the 1m candle action before deciding.
- **HOLD -- momentum intact** — no adverse signals; position is behaving as expected.

Example: You're in a long and see **EXIT -- microstructure deterioration (OFI, CVD, TFI)**. Three flow signals flipped against you at once — this is a fast exit signal, not a "wait and see" situation. Hit the market.

2. ATR Entry Levels

Now structure-first (v51). These levels are placed from **market structure first** — a confirmed swing, an HVN wall, or the POC — and only fall back to ATR multiples when no structural level fits. That makes them much closer to how you actually trade, but they're still **advisory**: the engine shows them, it doesn't place orders. The one gap that remains is the stop — see **STOP_CLAMPED** below.

ATR Entry Levels

What it shows: A structure-first trade frame — where the stop, entry, and target sit, with the target placed from the nearest relevant structure (falling back to an ATR multiple only when none fits).

What to watch for:

- The ATR value tells you how much BTC moves per bar on average right now. A higher ATR = wider, more volatile market.
- The header shows the fallback multipliers, with the target one **session-resolved** — 1.75x target on NY, 2.00x on London, 1.25x on Asia (v51). They only apply when structure doesn't place the level.
- size $\times N$ is your own sizing formula ($\text{Base} \times \text{AvgATR} / \text{CurrATR}$) rendered inline — above 1.0x means current ATR is *below* its rolling average (size up within risk limits); below 1.0x means current ATR is elevated (size down).
- EXEC $\langle N \rangle m$ tags the execution resolution — EXEC 1m for NY, EXEC 3m for Asia/London (v36, NY trades faster so it stays on 1-minute bars; Asia/London's typically lower volatility runs on 3-minute bars to keep moves tradeable). ATR and the levels are computed at that resolution — don't compare a 3m reading directly against a 1m one.
- **The placed-target row** renders in the arrow form when structure places it: Target $\langle \text{fallback} \rangle \rightarrow \langle \text{placed} \rangle$ [PLACED @ $\langle \text{placed} \rangle$ ($\langle \text{LABEL} \rangle$)] — and that row carries no inline R:R (read the structural rows below the ATR block for the swing R:R). The label says what the target is: SWING_HIGH_5M / SWING_LOW_5M (a swing), NEAREST_HVN_ABOVE / NEAREST_HVN_BELOW (a volume wall), POC, or FALLBACK_ATR (no structure fit — pure ATR, printed with the true R:R inline). A structural target can sit *farther* than the ATR fallback, not just closer — that's the v51 change.
- **The stop label** matters for risk: SWING_STOP = your real structural invalidation is being shown; STOP_CLAMPED = the structural stop was wider than $1.6 \times \text{ATR}$ so the engine tightened it to that ceiling (**common at current fixed sizing** — the shown stop is *tighter* than your true swing stop, so size accordingly); FALLBACK_ATR = no swing stop available, pure ATR.
- The R:R shown is at the **placed** levels — realistic, not a fixed config ratio. Fallback-only rows run ~1:1.1; structural rows are usually better.

Example: ATR ENTRY LEVELS (ATR 24.60 size $\times 1.89$ | 1.6x stop / 2.00x target | EXEC 3m) — a London 3-min session (hence the 2.00x session-resolved fallback target). Current ATR is well below its rolling reference, so size $\times 1.89$ says you can run nearly double size within your risk rules. If the target row shows Target 59284.2 $\rightarrow$ 59333.4 [PLACED @ 59333.4 (NEAREST_HVN_ABOVE)], the engine has placed the target at the nearest HVN wall above instead of the 49-point ATR fallback. If the stop reads [STOP_CLAMPED], your true swing invalidation is wider than the $1.6 \times \text{ATR}$ the engine is showing — don't mistake the tighter displayed stop for your real risk.

3. Kelly Sizing

Always advisory. The Kelly block uses a theoretical win probability, not measured results. Read it as “the engine's directional conviction” not as a position-sizing prescription.

Kelly Sizing

What it shows: A suggested risk fraction and contract count, translated from the engine's confidence level into sizing language.

What to watch for:

- [CAPPED] in the header means the engine thinks you have more edge than the 5% safety cap allows. Do not try to override the cap — the underlying probabilities are pre-calibration estimates, not measured win rates.
- [BIAS ONLY – NO TRADE] means the engine computed a size but the verdict is NO TRADE. The number is directional colour only — do not trade it.

- **< 1 contract (stop too wide for min size)** means the current volatility is so high that your full risk budget doesn't cover the minimum contract at this stop distance. The trade is impractical at current sizing.
- The **p(win)** displayed (45–65%) is a rough prior based on confidence tier, not a backtested statistic. Treat it with appropriate scepticism.
- A **Notional**: $\approx \$N \cdot N.N \times \text{lev}$ line follows the contract count — this is the dollar size and implied leverage the suggested contracts represent on a Deribit inverse contract. **[LEV CAPPED]** means leverage, not the dollar risk cap, was the binding constraint (**kelly.max_leverage**, default 5.0×) — the engine would otherwise have sized more contracts than the leverage ceiling allows.

Example: CONFIDENCE: HIGH → p(win): 65% → Applied fraction: 5.00% [CAPPED] → Contracts: 3 → Notional: $\approx \$30 \cdot 3.0 \times \text{lev}$. The engine is at its maximum conviction. The **[CAPPED]** tag means it would suggest more if it could. Scale the Risk \$ to your real account size (displayed value assumes a placeholder account).

4. Dynamic Norms

What it shows: How the engine has adapted its signal thresholds to current session conditions — volume, VWAP deviation, and ATR — rather than using fixed static values.

Dynamic Norms

What to watch for:

- **[LIVE]** tag = the engine is using fresh, adaptive thresholds. Normal operating state.
- **[STATIC FALLBACK]** tag = the engine couldn't compute fresh thresholds — likely a data issue. Treat the verdict with suspicion until the next run returns **[LIVE]**.
- **Vol threshold H and M:** H is the “high volume” bar; a candle must hit this ratio to fire a full volume signal. If H is low (near 2x), even modest volume spikes register — don't over-weight them. If H is high (4–5x), only genuine institutional-scale moves fire.
- **ATR ratio** (relabelled from “ATR scale”): current ATR vs its rolling reference. Below 1.0x = quiet relative to baseline; above 1.0x = expanding. This is now a **sizing-context figure only** — it no longer stretches or compresses the ATR stop/target distances in the entry block (those are a flat **ATR × multiplier** since the v32 display pass). Use it the same way as the ATR Entry Levels **size × N** figure: low ratio = size up viable, high ratio = size down.
- ATR is resolution-dependent since v36 (1-min on NY, 3-min on Asia/London) — current reference bands: **1-min** Low<20 / Normal 20–55 / High>55; **3-min** Low<42 / Normal 42–115 / High>115. These move with BTC's price regime — treat as a current read, not a permanent constant.

Example: ATR ratio: 0.79x with ATR=57.60, ref=72.58 — current volatility is 21% below the rolling average. Stop/target distances in the entry block are unaffected by this number; it's purely a “how hot is the market right now” read for your own sizing.

5. Regime

What it shows: The broader market structure state on the 5-minute chart — whether the market is trending, ranging, or in transition between the two.

Regime

What to watch for:

- **TRENDING_UP** — the engine only scores long signals with full weight. A short verdict in this regime is blocked automatically.
- **TRENDING_DOWN** — the engine only scores short signals with full weight. A long verdict is blocked.
- **RANGE_BOUND** — both directions are valid; mean-reversion signals get extra weight.
- **TRANSITIONAL** — the regime is ambiguous; the engine applies a scoring penalty (1–2 points) to dampen conviction. Size down in this state.
- Watch the +DI vs –DI spread: a large gap (e.g., +DI 27 vs –DI 9) shows strong directional bias even if ADX hasn’t crossed the trend threshold yet. This is the “direction is clean but the trend hasn’t been confirmed yet” read.

Example: REGIME: TRANSITIONAL | ADX: 24.3 | +DI: 27.5 | –DI: 9.6 — ADX is just below the trending threshold of 25, but +DI dominates strongly. Watch for ADX to push through 25 over the next few runs; if it does, the regime flips to **TRENDING_UP** and long verdicts get a scoring boost. For now, treat any long verdict with modest conviction and size conservatively.

6. Core Signals (1m)

ROC(9) — Rate of Change

What it shows: How much price has moved in percentage terms over the last 9 candles, and whether that movement is accelerating, flat, or reversing.

What to watch for:

- Positive ROC + **RISING** slope = clean bullish momentum; the engine scores this as a full long signal.
- Positive ROC + **FLAT** slope = still positive but losing steam; partial score only — watch for confirmation from other signals.
- $|ROC| > 0.3\%$ on a 1-minute candle is a strong single-bar impulse. $|ROC| > 0.6\%$ is extension territory.
- ROC crossing zero while you’re in a position is a structural exit trigger — the engine will flag this in the HOLD/EXIT line.
- On Asia/London (3-min execution since v36), the magnitude and slope thresholds that gate a full ROC signal are re-baselined per session (v40/v41): ASIA fires at 0.17, LONDON at 0.11 — Asia genuinely runs hotter on 3-min bars, so the same raw ROC reading means something different by session. NY (1-min) is unaffected.

Example: ROC(9): 0.115 | Slope: FLAT — price is marginally positive but momentum is stalling. The engine gives this a partial long score only. Paired with **MOMENTUM_FADING** context, this is a signal to stay out rather than buy.

RSI(9)

What it shows: Whether price is in overbought or oversold territory on the 1-minute chart, and whether a momentum divergence (price and RSI disagreeing) has been detected.

What to watch for:

- RSI above 60 = full long signal. RSI below 40 = full short signal.
- The 45–55 zone is a dead band — RSI here provides no directional edge and the engine scores it as neutral.
- Div: **BEARISH** shown with RSI between 58–64 = soft warning to tighten exits, but not a scored penalty. Only triggers a score penalty when RSI is above 65.
- Use RSI for **trade management**, not entry: while in a long, RSI > 60 = hold, RSI < 40 = exit.

Example: RSI(9): 52.6 | Div: BEARISH — RSI is in the dead band (no directional score), and there’s a divergence flag, but RSI is too low for the penalty to trigger. Read it as a soft caution: momentum is not confirming direction, and the divergence is a background warning. Not a trade entry signal in either direction.

Volume

What it shows: The current 1-minute candle's volume in BTC and USD, how it compares to the recent average (SMA), and whether it's high enough to confirm a directional move.

What to watch for:

- Volume ratio at or above the H threshold (shown in Dynamic Norms) = strong volume confirmation; fires a full volume score when direction and VWAP position also agree.
- Volume between M and H thresholds = mid-tier; the engine requires additional confirmation before scoring it.
- Volume below 0.7x (shown in red) during price movement = fade warning. Price moving without volume rarely sustains; treat the move with suspicion.
- The USD notional column matters: a low BTC ratio can still represent meaningful participation if the dollar value is large (e.g., > \$500K on a 1m candle).

Example: Volume: 0.37 BTC (\$28.1K) | vs SMA: 0.16x — this is extremely low participation. No volume signal fires. Any price movement on this candle is low-conviction; don't chase it.

7. VWAP

VWAP (Volume Weighted Average Price) — the average price for the session, weighted by how much volume traded at each level. It represents the market's "fair value" for the day.

VWAP

What it shows: Where price sits relative to the session's fair value, and how far it has stretched from the average.

What to watch for:

- [WARMUP] tag = the session just reset (00:00 or 13:30 UTC); VWAP scores are suppressed for the first ~15 candles. Don't trust VWAP-based signals until the tag disappears.
- Price between VWAP and the s1 band = the engine's cleanest VWAP signal zone; full score fires here.
- Price between s1 and s2 = extended; partial score only, requires cross-confirmation.
- Price beyond s2 = too stretched for a fresh-direction signal in either direction; the engine scores nothing here by design.
- Dev > 0.3% = meaningful deviation from fair value. Dev > 0.5% = extended, often mean-reverts.

Example: Value: 75995.6 | Dev: 0.056% | s1 band: [75715.9, 76275.3] — price is sitting 0.056% above VWAP, well within the s1 band. This is the ideal "price hugging fair value" posture for a long setup. The engine can fire a full VWAP long score here with no stretching penalty.

8. BBW / TTM Squeeze

Bollinger Band Width (BBW) measures how much price is compressing. A squeeze means the market is coiling energy before a move. **TTM Squeeze Momentum** shows which direction that energy is pointing.

BBW (Bollinger Band Width)

What it shows: Whether the market is in a volatility squeeze (energy coiling) or releasing from one.

What to watch for:

- **Status: ACTIVE** = market is in the bottom 20% of recent volatility; a breakout is building. Don't fade moves in this state.
- **Status: RELEASING** = the squeeze just fired; a directional move is in progress. The engine boosts signal scores in this state.
- **Status: NONE** = normal volatility conditions; no compression signal.
- Use BBW alongside TTM Direction — a **RELEASING** BBW with a clear TTM direction is the engine's preferred breakout posture.

Example: BBW: 0.004 | Status: NONE — no squeeze currently active. This is a neutral read; normal scoring applies without any squeeze boost.

TTM Squeeze Momentum

What it shows: The directional momentum reading from the TTM Squeeze oscillator — whether bullish or bearish momentum is growing or fading.

What to watch for:

- **Signal: BULL_MOMENTUM** or **BEAR_MOMENTUM** = momentum is building in that direction; full score contribution.
- **Signal: BULL_FADING** or **BEAR_FADING** = momentum exists but is decelerating; partial score only, and contributes to a **MOMENTUM_FADING** context tag.
- **Dir: RISING** or **FALLING** tells you which way the histogram bars are moving — use this to confirm the signal.
- A histogram turning from positive to negative (or vice versa) is a momentum crossover; watch for it across consecutive runs.

Example: Histogram=70.26 | Dir=FALLING | Signal=BULL_FADING — bullish momentum is present (positive histogram) but the bars are shrinking. The engine treats this as a partial long contribution only, and it will add weight to any **MOMENTUM_FADING** context tag. Not the time to add to a long position.

9. EMA Ribbon

EMA (Exponential Moving Average) — a moving average that weights recent prices more heavily. A “ribbon” uses multiple EMAs of different lengths to show trend alignment across timeframes.

EMA Ribbon

What it shows: Whether the short-term, medium-term, and long-term trend are all pointing in the same direction.

What to watch for:

- **BULL** = all EMA layers are aligned upward; full long score.
- **BEAR** = all layers aligned downward; full short score.
- **BULL_WEAK** / **BEAR_WEAK** = partial alignment; one or more layers are out of sync. Partial score only.
- **NEUTRAL** = no alignment; no score contribution.
- Check the 5m EMA200 line separately — if price is above the 200-period EMA on the 5-minute chart, the macro trend is bullish; below it is bearish. This gate can block signals that go against the macro direction.

Example: EMA Ribbon: BULL_WEAK | 5m EMA200: Price above — short-term trend is mostly aligned bullish but not fully stacked. The macro backdrop (price above 5m EMA200) supports the long direction. A partial score contribution; keep watching for the ribbon to fully align.

10. Market Structure

Market Structure

What it shows: The pattern of swing highs and lows, whether price is breaking out of a range, and whether volume is confirming the trend direction.

What to watch for:

- **Donchian breakout:** Price breaking above the recent range high = bullish breakout signal. Breaking below the range low = bearish. No break = neutral, no score.
- **OBV (On-Balance Volume):** OBV — the running total of volume flowing in and out — should trend in the same direction as price. OBV rising while price rises = healthy trend. OBV diverging from price (one going up, the other down) = warning signal; the engine can flag this as an exit trigger for open positions.
- Watch for OBV divergence in the HOLD/EXIT line if you're already in a trade.

Example: Price makes a new high above the Donchian channel but OBV is flat or declining. The engine flags this as a structural divergence. Don't add to longs here — the volume isn't backing the breakout.

Swing Pivots (5m + 15m)

What it shows: The most recent confirmed swing high and swing low on both 5m and 15m candles. Drives the structural stop and target lines under the ATR Entry Levels block, and the Layer 1.5 structural-break exit in the HOLD/EXIT line.

What to watch for:

- The 5m swing low under your long entry is your structural stop — price breaking it cleanly means the long premise is invalidated. The engine fires EXIT -- **structural break (swing low breach)** when this happens.
- The 5m swing high above your long entry is your structural target. R:R is the ratio printed in the structural row of the entry block.
- 15m swings are context only — they don't drive stops/targets but tell you whether the broader structure agrees. A 5m setup against the 15m swing direction is lower-quality.
- A STRUCTURALLY_WEAK context tag fires when swing data exists but no clean target+stop pair can be placed for the verdict direction.

Example: LONG Stop: 71450 Entry: 71850 Target: 72420 R:R 1.4 reading from the structural row — the 5m swing low at 71450 is your stop, swing high at 72420 is your target. If price drops below 71450 mid-trade, the engine flags EXIT -- **structural break** regardless of what the indicators say.

Trend Structure (HH/HL/LH/LL)

What it shows: The classification of the last six confirmed 5m swing pivots into one of five patterns. This is a Pass 2c scoring input — the structure bonus pushes the dominant side of a confirmed-direction setup.

What to watch for:

- UPTREND (HH+HL — higher high, higher low) — adds +1 to long score when long is dominant. This is the cleanest bull structure read.
- DOWNTREND (LH+LL) — adds +1 to short score when short is dominant.
- EXPANSION (HH+LL — both higher high and lower low) — range-widening, no scoring change. Treat as a caution flag — the market is in directional disagreement at higher and lower extremes simultaneously.
- CONTRACTION (LH+HL — narrowing range) — no scoring change. Often precedes a break; pair with BBW squeeze for breakout posture confirmation.
- UNDEFINED — fewer than 2 highs and 2 lows in the lookback. Insufficient structure to classify.

Example: Trend Structure: UPTREND (HH 102450.0 > 102100.0 | HL 101800.0 > 101500.0) — both the most recent high and most recent low are higher than their priors. A LONG verdict in this state gets the +1 bonus. A SHORT verdict in this state would not — the structure disagrees with the direction.

Best Volume Pivot (Display-Only)

What it shows: The pivot in the 5m lookback with the highest total volume across its wing window, plus the volume ratio against the average pivot. Not currently used in scoring or cap arbitration — treat as a chart-reading aid.

What to watch for:

- **Best Vol Pivot 5m:** HIGH 102450.0 (vol×2.3 vs avg pivot) — the 102450 swing high was made on 2.3× the average pivot volume. That's a stronger structural reference than a same-price level made on average volume.
- When the “best” pivot price differs from the most-recent swing being used as your target/stop, eyeball whether it's a stronger structural level worth referencing for partials.
- Ratios above 2.0× are meaningful; under 1.5× the volume-weighting is barely differentiating from the most-recent pivot.

There is a switch for this, and it is off. Since v63 the engine *can* place its target on this pivot instead of the most-recent swing — but the setting that does it (use_best_pivot_candidate) ships **off**, and nothing on your screen changes while it is off. It was built so the idea could be **backtested before anyone turns it on live**, not to change how the engine behaves. Until that study says otherwise, the pivot on this line is something for *you* to read, not something the engine acts on.

If you do run the sweep — What-If Replay → **Use best-pivot candidate, 0:1:1**, in this guide's §17 *Working with the App* — this is what switching it on does:

- The pivot becomes the **first** choice of target, ahead of the swing, whenever it qualifies. It is a target candidate only; your stop is unaffected.
 - It qualifies on **which side of entry the price sits**, not on whether it was a high or a low. A *low* pivot above entry still counts as a long target — it is a level the market defended on volume, and that is the part that matters for placement.
 - It has to sit within the same reach bound as every other structural target (Target max ×ATR), so a far-away pivot is ignored rather than stretched to.
 - Because it moves where targets land, it can also pull some trades **under** the min-move floor and turn them into BELOW_MIN_MOVE — read the population-shift line, not just the hit rate.
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11. Open Interest

Open Interest (OI) — the total number of active contracts currently open in the market. Rising OI means new money is entering; falling OI means positions are being closed.

Open Interest

What it shows: Whether new capital is flowing into the market in the direction of the current move, or whether the move is being driven by short covering and position closing.

What to watch for:

- Rising OI + rising price = bulls opening new longs; strong, conviction-backed up move.
- Rising OI + falling price = bears opening new shorts; strong, conviction-backed down move.
- Falling OI + rising price = shorts closing (short covering), not new longs. This move has less staying power.
- Falling OI + falling price = longs closing (long liquidation). Weaker move.

- The engine cross-references OI delta with CVD (see Order Flow) for a combined conviction signal; both agreeing produces a scoring boost.

Example: OI Delta: BULL | CVD: BULL — both new money and the actual buy/sell flow are confirming the long side. This is the engine's strongest OI confirmation state and triggers a score boost above a single-signal read.

12. Order Flow

Order Flow shows the real-time balance between buyers and sellers based on how orders are actually hitting the market. It tells you who is in control right now.

Order Flow

What it shows: Three flow signals — OFI, CVD, and TFI — each measuring buyer/seller dominance from a different angle.

What to watch for:

- **OFI (Order Flow Imbalance):** Measures the imbalance between buy and sell orders sitting on the bid vs ask. Positive = buyers dominating. Since v46, on the live WebSocket path the ratio is a **time-weighted average** over the run window (not a single snapshot) — a brief sweep or spoof can no longer flip it; what you're reading is sustained imbalance, not an instant. (Falls back to a single snapshot when running on REST.) **Cosmetic note:** because it's an average-of-ratios rather than a ratio-of-averages, the displayed Bid Vol / Ask Vol won't always divide out to exactly the displayed ratio — that's expected, not a bug.
- **CVD (Cumulative Volume Delta):** Running tally of buy volume minus sell volume over the session. Positive and rising = sustained buying pressure. Pay attention to the **MicroCVD** sub-reading — if it shows BULL_DECEL, the recent buying is slowing down even if the cumulative total is still positive.
- **TFI (Trade Flow Imbalance):** Similar to CVD but measured over a shorter, more recent window. Useful for catching rapid flow reversals.
- All three agreeing (e.g., OFI: BULL, CVD: BULL, TFI: BULL) = the engine's cleanest order flow confirmation.
- Two or three of these flipping against your open position simultaneously triggers the EXIT -- microstructure deterioration alert in the HOLD/EXIT line — that's a fast exit signal.

Example: OFI: BULL | CVD: BULL | TFI: BEAR — two of three confirm bulls, but TFI (the most recent window) just flipped bearish. Not enough to exit if you're in a long, but worth watching. If OFI also flips on the next run, you have a microstructure deterioration developing.

OFI Momentum

What it shows: Whether the OFI level signal is accelerating or fading over the last few runs. RISING/FALLING/FLAT, computed against a 10-sample ring buffer.

What to watch for:

- OFI: BULL with OFI Momentum: RISING — the bullish flow is intensifying. The engine adds a momentum modifier to the OFI level score.
- OFI: BULL with OFI Momentum: FALLING — the bullish read is technically still there but losing force. Lower-quality long signal.
- FLAT momentum on a strong OFI level is normal — it just means the imbalance has been stable. Not negative.
- A momentum shift (RISING → FALLING in consecutive runs) on a position you're holding is an early warning the flow is rolling over — earlier than waiting for the level signal itself to flip.

Example: OFI: BULL | OFI Momentum: RISING on the same run as a LONG verdict — the buy-side flow is accelerating, not just present. This is the cleanest fresh-entry posture for a long; flow is not just leaning bullish but doing so harder than a few runs ago.

Bid-Ask Spread (WIDE penalty)

What it shows: The current bid-ask spread in basis points. When the spread is unusually wide, the engine applies an entry-side penalty — the assumption is that the order book has thinned out, often during a flush or news event, and is not a clean entry environment.

What to watch for:

- A Spread WIDE flag in the breakdown table means a penalty was applied to the verdict-side score. The verdict may degrade by one tier as a result.
 - Wide spread + apparent directional signal often means the move is already half-finished by the time the next candle prints. The engine is biasing toward “no trade” rather than chasing.
 - During normal market conditions on BTC-PERPETUAL, the spread sits in the 1–3 bps range. Anything above 5 bps is treated as wide.
 - Don’t second-guess this penalty — it specifically catches the “looks like a great signal but the book is empty” trap.
-

13. Liquidations

What it shows: Recent forced position closures (liquidations) and which direction they’re coming from — bullish liquidations (shorts being wiped out) or bearish liquidations (longs being wiped out).

What to watch for:

- A cluster of bearish liquidations (long liquidations) during a down move = forced selling is adding fuel to the drop, but also potentially nearing exhaustion if the cluster is large.
- A cluster of bullish liquidations (short liquidations) during an up move = short squeeze fuel; the move can accelerate rapidly.
- Liquidations after an extended move (rather than at the start) can signal exhaustion — the weak hands have been cleared.
- The engine uses liquidation data to add context to directional signals; a move backed by liquidations on the opposing side is treated as stronger.

Example: Price drops sharply and you see a large cluster of long liquidations flagged. This is cascading forced selling — the move has real momentum behind it. But if this comes after a 30-minute sustained drop, the liquidation pool may be thinning and a bounce is more likely.

14. MTF Gate

MTF = Multi-TimeFrame. The MTF Gate checks whether the 15-minute chart agrees with the 1-minute signal. It acts as a hard filter — even a perfect 1m setup gets blocked if the 15m disagrees.

MTF Gate

What it shows: Whether the 15-minute timeframe is aligned with the 1-minute verdict. If it isn’t, the engine can block the trade entirely.

What to watch for:

- **PASS** = 15m is aligned with the 1m direction; the gate is open, verdict stands.
- **BLOCK** = 15m disagrees; the verdict is overridden to **NO TRADE [WEAK X]** regardless of how clean the 1m setup looks. Respect this block — counter-trend scalping on 1m against 15m structure has poor expectancy.
- The 15m gate uses EMA alignment, trend direction, and momentum on the higher timeframe — it's not a single indicator.
- A **BLOCK** that clears on the next run (15m catches up) is the ideal entry sequence: wait for the gate to open, then enter on the next confirmed 1m signal.

Example: VERDICT: **STRONG LONG** with MTF Gate: **BLOCK** → output reads **NO TRADE [WEAK LONG]**. The 1m setup is clean but the 15m is still pointing down. Stand aside. When the MTF Gate flips to **PASS** and the next 1m run still shows a long verdict, that's your entry window.

15. Funding

Funding rate — in perpetual futures, traders on the long side pay traders on the short side (or vice versa) every 8 hours. A high positive funding rate means longs are paying shorts; high negative means shorts are paying longs.

Funding

What it shows: Whether the current funding rate is aligned with, neutral to, or working against your trade direction — and whether it is significant enough to affect scoring.

What to watch for:

- **Positive funding (longs pay shorts):** The market is leaning long. Slightly bearish contrarian signal — longs are crowded. The engine may reduce the long score or boost the short score.
- **Negative funding (shorts pay longs):** The market is leaning short. Slightly bullish contrarian signal — shorts are crowded.
- **Near-zero funding:** Neutral; no impact on scoring.
- Extreme funding (very high positive or negative) is a stronger signal; the engine scales the modifier based on magnitude.
- Funding is most relevant for **trade bias** on holds — if you're in a long and funding is sharply positive, you're paying to hold and the crowd is with you, which means a reversal could be sharp.

Example: Funding: **+0.025% (HIGH)** — longs are paying a high rate to hold. The engine applies a modest penalty to long scores as a contrarian signal. This doesn't block a long trade but is a reminder that the long side is crowded. If you enter long, keep your target tighter than usual.

Funding Momentum

What it shows: The direction of the funding rate change over the last 3 samples (≈3 minutes at 60s polling). **RISING / FALLING / FLAT.** Acts as an adjunct to the absolute funding signal in Step 3b — amplifies or softens the penalty depending on whether the crowding is intensifying or unwinding.

What to watch for:

- Funding: **+0.025% (HIGH)** with Funding Momentum: **RISING** — long crowd is still piling in. The engine amplifies the long-side penalty. Avoid late long entries.
- Funding: **+0.025% (HIGH)** with Funding Momentum: **FALLING** — crowd is unwinding even though the absolute rate is still elevated. Penalty is softened. Short window for a long re-entry if other signals support it.
- **FLAT** is the most common state at any given minute — funding moves slowly relative to 1m candles.

- Like the absolute rate, this is a confidence modifier — it does not block a trade, only nudges score weight.

Example: Funding: -0.018% (LOW) | Funding Momentum: FALLING — shorts are leaning the market and the short-side rate is becoming more negative (i.e., shorts paying more). Combined, this is a contrarian setup for longs: the short crowd is intensifying its position, classic squeeze-fuel posture if a long signal is otherwise present.

16. Signal Breakdown Table

What it shows: The full itemised score ledger — every indicator that scored, what it scored ([L] for long, [S] for short, [L*] / [S*] for partial, or a penalty), and the running total that matches the SCORE line in the Verdict Block.

What to watch for:

- The TOTAL row at the bottom must match the SCORE line in the Verdict Block. If it doesn't, the run may have had a data issue.
- Look at the balance: are most [L] hits from structural indicators (VWAP, EMA, Donchian) or from flow indicators (OFI, CVD, OI)? A verdict supported by both is stronger than one supported by just one category.
- Penalty lines (shown as negative scores) are as important as positive hits — they tell you what the engine is worried about.
- [L*] / [S*] = partial scores that didn't get upgraded. A verdict with many partials and few full hits is weaker than a verdict with mostly full hits, even at the same total score.
- Use this table to understand **why** the engine called what it called — not just what it called.

Example: VERDICT: LONG | SCORE: Long 11/20 — looking at the breakdown, 7 of those 11 points are from flow signals (CVD, OFI, TFI, OI Delta) and only 4 from structure (VWAP, EMA). This is a flow-heavy long. It can work, but is more vulnerable to a sudden flow reversal than a structure-heavy verdict with the same score. Consider a tighter stop.

17. Working with the App

These features sit alongside the per-run output above. They affect how the engine adapts over time and how you investigate its accuracy.

Analysis Report Viewer

What it shows: A markdown report joining recent verdicts to subsequent price action — failure rate per verdict tier, hold-window stability, funding momentum diagnostic, OFI outlier audit, OI×CVD asymmetry breakdown.

Where: Status bar at the bottom of the main window, link labelled **Analysis Report**.

What to watch for:

- **“Hold Window Selection Stats”** — answers the practical question: “STRONG LONG verdicts have been most reliable held for how long?” Per-tier table with the empirically chosen window. Use this to set your default scale-out timing per tier.
- **Failure-rate matrix** — per-tier × window × ATR threshold cells with sample size and 95% confidence interval. Cells with $n < 30$ are flagged as insufficient sample; ignore them until more data accumulates.
- **Funding Momentum Diagnostic** — distribution of raw funding-rate deltas, percentile table. Tells you whether the FLAT-heavy FundingMomentum classification is genuine quiet vs threshold mismatch.
- **OFI Outlier audit** — flags rows where OFI Ratio exceeded 100 or 1000. If you see a recurring outlier pattern, raise it as a possible calculation issue.

When to read it: After ~500 new rows accumulate, or when verdict accuracy seems off. The auto-tweaker reads the same data programmatically, but the markdown report is the human-readable view.

Tweak Settings — The Auto-Tweaker

What it does: Periodically reviews recent verdict accuracy (failure rate against tiered ATR-adjusted forward returns), and when accuracy slips below a threshold, asks Claude Opus to propose targeted settings adjustments. Settings get applied either automatically or after your manual approval, depending on the toggle.

Where: Status bar link labelled `Tweak Settings` — opens a non-modal dialog.

What you control:

- **Auto-commit on apply** — when on, accepted settings diffs are written to `settings.json` directly. When off, the diff is parked at `tools/AutoTweaker/proposed_diffs/` for you to review and apply manually.
- **Dry-run mode** — when on, the auto-tweaker generates the API call payload as a `.txt` file at `tools/AutoTweaker/dry_run_payloads/` and stops. No API call is made. Useful for testing without burning API credits.
- **Window size (verdicts)** — number of recent verdicts to evaluate failure rate over. Default 120. Must all fall within the same UTC session.
- **Failure rate threshold (%)** — if aggregate failure rate exceeds this, a tweak is proposed. Default 40%.
- **Cooldown rows** — minimum new verdicts between tweak attempts. Default 10.

Status indicator:

- **Ready** — all conditions met, “Run Tweaker Now” enabled.
- **Cooldown: N rows remaining** — too few new rows since last run.
- **Waiting for session-aligned window: M/120 rows** — current session hasn’t accumulated enough rows yet.
- **Insufficient tier-eligible rows** — too many of the recent verdicts were `NO TRADE` or `WEAK_*`; not enough `STRONG/MEDIUM` verdicts to compute a reliable failure rate.

What it will not do:

- Touch any rejected pattern (fixed-% targets, double-counting setups, dead v15 keys). The applier hard-rejects these regardless of what Claude proposes.
- Disable the MTF gate or the regime-weights gate. Hard-coded blocks.
- Change more than 3 keys per proposal. The 3-key cap is the conservative-bias safeguard.
- Touch anything you own — your risk sizing (`kelly.*`), the per-session / per-resolution re-baseline keys (`resolution_profiles.*`, `session_volume.sessions[].*`, the `aggressor_velocity` and `structural_levels` session overrides), or the feature switches and display/ops blocks. The applier hard-rejects them.

Which knobs are yours vs the tweaker’s. Every `settings.json` key sits in one of three ownership tiers — auto-tweaker-tunable, hand-tuned re-baseline, or hand-toggle switch. The full per-key table is in **User Manual §19** (“Settings ownership tiers”). Short version: the tweaker only moves failure-rate thresholds; anything session-specific, or any on/off switch, is yours.

Reading the “Last Run” line:

- `[BELOW_THRESHOLD]` — engine is performing fine, no tweak needed. This is the normal happy state.
- `[INELIGIBLE]` — cooldown / session-not-aligned / insufficient tiers. No action. Not an error.
- `[DRY_RUN_WRITTEN]` — payload file generated, awaiting your manual handling.
- `[PROPOSED]` — diff parked, awaiting manual apply.
- `[APPLIED]` — settings updated, version bumped, `change_log` appended.
- `[ERROR]` — API call or validation failure. See the summary line for the reason.

The auto-tweaker is a long-loop optimiser. Don't expect frequent tweaks — most of the time, the verdict accuracy is fine and the run reports `BELOW_THRESHOLD`. When a tweak does fire, treat it as a calibration event worth reviewing rather than a routine update.

Settings Snapshots

When the auto-tweaker has run for several consecutive rounds without proposing any settings changes, the engine considers those settings “proven” for the current market conditions and saves a snapshot. If market conditions later revert to a similar pattern, the auto-tweaker may propose reverting to one of these snapshots instead of tweaking fresh keys.

You can see:

- **Active snapshot** (if any) and the running streak counter in the Tweak Settings dialog.
- **Full history of saved snapshots** in `settings_snapshots/manifest.csv`. The directory is gitignored.
- **Round-level statistics** for the last 5 rounds via the **Show Round Stats** button inside the Tweak Settings dialog. Each round shows the aggregate failure rate plus per-tier accuracy (`STRONG` / `MEDIUM` / `WEAK` on both sides; `NO_TRADE` rows are informational only).
- **Open Snapshots Folder** button in the Tweak Settings dialog opens the snapshots directory in your file explorer.

How snapshots are scored. Snapshots are bucketed by `regime × volatility tier` (12 buckets total). Only one snapshot is kept active per bucket — when a new finalised snapshot scores higher than the existing one in its bucket, the older one is rotated and its `.json` file is deleted (the manifest row remains as a historical record). Score blends failure rate (weighted heavier — 1 point per percentage point) and streak length (1.5 per round up to 20).

When a revert proposal fires, it follows the same `auto-commit` / `dry-run` toggles as a regular tweak. The revert is a wholesale settings replacement, so the per-proposal key cap does not apply — but the snapshot's content is still validated against the rejected-pattern list before any apply.

Snapshot-related knobs in Tweak Settings:

- **Snapshot streak X** (default 3) — how many consecutive `BELOW_THRESHOLD` rounds before a snapshot is saved.
- **Max keys per tweak proposal** (default 3) — previously hard-coded; the conservative-bias cap that limits how aggressive a tweak can be. Reverts are exempt by design.
- **Streak weight** (default 1.5) — composite-score weight per streak round, capped at 20 rounds.

Live Performance Strip

A compact strip of six labels updates after every analysis run showing how the engine's directional verdicts have been performing over multiple time windows. Use it to gauge whether the current settings are working across the session or if the recent batch of signals has been noise.

What each label shows:

Label	Window
Cur.Wk	Monday 00:00 UTC+8 → now
3d	D-2 00:00 UTC+8 → now (last three calendar days)
Cur.Day	Today 00:00 UTC+8 → now
Asia	Most-recent Asia session block (08:00–15:00 UTC+8)
London	Most-recent London session block (16:00–20:00 UTC+8)
NY	Most-recent NY session block (21:00–07:00 UTC+8 next day)

Reading the colours:

- **Green** — success rate > 50% in this window. The engine has been calling direction correctly more often than not.
- **Red** — rate ≤ 50%. Signals have been losing or breaking even; tighten style or reduce size.
- **--%** — fewer than 4 evaluated predictions in the window. Not enough data to trust the rate.

Hover tooltip on any label shows the exact sample count and time range, e.g. **12 predictions evaluated. 2026-05-12 08:00 → 2026-05-13 10:45 UTC+8.**

Session “most-recent block” semantics. Each session label always shows the most-recently active or completed block for that session — not a rolling 24h average. If you look before today’s Asia session has started, the label covers yesterday’s Asia block (fully completed). Once Asia opens, it switches to today’s running block and grows. This means session rates are directly comparable to your own session-level P&L.

Success metric. A verdict is counted as SUCCESS if price reached the target barrier before hitting the stop barrier, evaluated on 1m bars T+3 through T+15 (1-min sessions; **3-min Asia/London rows walk T+3 through T+45** — the same windows in bars, scaled post-v41 — so their outcomes show up ~45 min later). Ties in the same bar count as FAILURE (conservative-bias rule). **NO TRADE** and **NO TRADE [WEAK X]** verdicts are not counted — only directional signals enter the denominator.

v51 caveat. The strip’s barriers use the placed target when one exists, else a $2.0 \times \text{ATR}$ fallback, and the *raw* swing stop else $1.2 \times \text{ATR}$. These eval multipliers were kept on the old yardstick when v51 moved the *displayed* levels to $1.75 \times / 1.6 \times$, so failure rates stay comparable across the change — but on a fallback row the strip scores against slightly wider barriers than the levels shown on screen. Bounded and known; a migration onto the logged placed levels is queued (D6).

Cold start. On first launch the strip shows --% while 7 days of OHLC history loads and the eval cache backfills. This takes a few seconds; the status bar shows “Loading performance history...” while it runs. Subsequent launches load from the cached files in under a second.

Settings knob. `performance_display.enabled = false` hides the strip and stops writing the two cache files. Use `min_sample_for_render` to raise the threshold before a rate number appears (default 4).

WebSocket Health

The engine has run live on a WebSocket feed since v42 (replacing REST polling as the primary data path). The status bar carries a health line so you can tell at a glance whether you’re looking at fresh data:

- **WS OK · 1/3/5/15 fresh · trades N** — all candle series fresh, normal operating state.
- **WS DEGRADED – REST fallback (stream stale)** — the WS stream went stale for this run; the engine fell back to REST for that single run rather than skip it. Occasional flickers are normal; persistent DEGRADED means the feed is struggling.
- **WS DOWN – reconnecting (Xs backoff, R reconnects)** — disconnected, retrying with exponential backoff. The app keeps functioning on REST fallback while down.

This is a live, in-memory status line, not part of the rendered verdict — it doesn’t get logged to CSV or the output dump. Check it if a run looks suspiciously stale or a verdict seems out of step with the tape.

Exit Guard

While you have a position declared (via the position radio buttons), a separate fast-cadence check runs every few seconds against the live WebSocket feed — independent of your normal auto-run interval — watching for the same fast-exit conditions as the HOLD/EXIT line’s microstructure layer (2+ adverse flow signals flipping at once, or a structural swing-level breach).

- **EXIT GUARD · clear** — no adverse condition present.

- EXIT GUARD · Δ EXIT? confirming n/d — an adverse condition is present but hasn't held for the debounce window yet (default 2 consecutive checks). Building toward a latch, not yet a signal.
- EXIT GUARD · Δ EXIT – <reason> — latched. An optional alarm sound fires on this transition. This is the same urgency as a HOLD/EXIT fast-exit line, just at tick freshness between full runs.

It's display/alert only — it never changes the verdict, never writes the CSV, and is paused (not silently stale) if the WS feed itself is down or cooling down. Requires `transport=ws`; shows "WS only" on REST.

On-Close Trigger Mode

By default the engine fires on a fixed interval timer. An alternative **ON-CLOSE** mode (toggle next to SINGLE/REPEAT) fires the analysis the instant the execution-resolution bar closes instead — NY's 1-minute bar, or Asia/London's 3-minute bar — so you get the verdict right at the structural decision point (bar close) rather than waiting up to a full interval for the next poll.

- The interval control relabels to **BACKSTOP** in this mode — it still fires if the bar-close watcher hasn't fired in that long (covers a stalled feed), but it's not the primary trigger.
- Next close: M:SS [SINGLE/REPEAT <res>m] counts down to the next bar boundary — the bracket echoes your SINGLE/REPEAT setting and the active resolution (1m NY, 3m Asia/London).
- Falls back to interval mode automatically if running on REST (`transport=rest`) — on-close detection needs the live WS bar stream.
- This only changes *when* a run fires, not what it computes — same verdict you'd get from a timer fire at that instant.

Live Microstructure Strip (TAPE)

A continuously-updating one-line strip (toggle: the **TAPE** checkbox) showing fast streaming microstructure *between* full analysis runs, refreshed every couple of seconds:

76038 · SL 75920 (–118) | SH 76210 (+172) · TFI BUY +0.42 · 1.8 bps · book 2.3× bid · 4.1 tr/s (\$312k/s)

- Last price, bracketed by the nearest structural level above and below (carried from the last full run's swing/VPFR levels — not recomputed live).
- TFI, spread, and top-book imbalance, refreshed live.
- Tape speed — trades/sec and \$/sec over a short rolling window.

This is deliberately not a signal. It's the same raw microstructure inputs the verdict pipeline uses, shown faster and rawer. The full verdict is still the considered, multi-indicator product — don't treat a TAPE reading as a trade trigger on its own; it's there so you're not flying blind on flow between full runs (e.g. while watching a level for entry, or managing a hold).

What-If Replay — backtesting a settings change

Where: SETTINGS & TOOLS → **What-If Replay**. The report opens in the Analysis Report Viewer.

Takes the runs already logged plus 1-minute price history, applies a hypothetical settings change, and re-computes — on the *same* historical rows — what levels the engine would have placed and what it would have called. Then it re-walks what price actually did and shows **current settings vs your hypothesis, side by side**. It answers "would a wider LONDON stop have helped on the book so far?" without waiting for another month of forward data.

It never changes anything. No `settings.json` write, no orders, no effect on the live verdict. A good result is evidence for a spec proposal — not a change.

Three inputs, that's the whole language. Per field: **blank** = leave it live · **one number** = pin it · **from:to:step** = sweep it. Sweep one knob to find its best value; sweep several and it runs every combination. A checkbox prefills the default sweep ranges so you can start from the full set and edit down. Cap 3,000 cells (the ranking shows the top 50). Note sweeps land on **from + k×step** — 1.5:2.0:0.2 gives 1.5/1.7/1.9, **skipping 2.0**.

Reading the result — in priority order:

1. **Holdout EV, not the headline.** Cells are ranked by EV per trade in ATR (not win-rate — a high win-rate with a tiny target still loses). The winner is picked on half the book and re-checked on the unseen half. **△ DIVERGENT means the edge vanished out-of-sample — discard it.**
2. **Population shift.** Did the change add or remove trades? A verdict-threshold change moves this; a stop/target change usually doesn't. If success went up because 14 marginal trades disappeared, that's survivorship, not skill.
3. **The matrix is the mechanism, EV is the bottom line.** A wider stop can improve the hit-rate table (fewer stop-outs) while *hurting* EV, because you lose more when you do get stopped.
4. **The overfit counter.** It tallies how many combinations you've tried against this book. Try enough and something will look good by chance — the header spells out how many false winners to expect.

Two knobs are worth knowing apart. *Stop max × ATR* is the clamp that binds most structural rows — the high-leverage stop lever. *ATR stop mult* is only the fallback, used where no structural stop exists, so sweeping it usually shows almost nothing. That's expected, not a broken run. (Likewise *Stop min floor ticks* is a guard that essentially never bites.)

Barriers are mid-price touches — no fills, no slippage, no queue. Real execution is worse than any number here.

Practical read: a result worth acting on has a **positive holdout EV, no DIVERGENT flag, a decent sample, and no shrunken population**. Everything else is a null result — which is still useful. Sweeps of the ATR fallback geometry on the current book keep coming back flat with divergent winners; that's the tool steering you off a phantom rather than handing you one.

BacktestRunner — the offline command-line tool

Where: not in the app. It's a separate console program (`tools/BacktestRunner/`), built with the solution and run from a terminal. There is no button for it.

Everything above works off runs the engine has *already logged*. BacktestRunner works off **raw market history** instead, so it can answer questions about periods the engine never watched — and it can check the tape itself for holes. It never writes settings, never places orders, and never touches the running app's log.

Five things it does:

Verb	What it's for
<code>fetch</code>	Downloads BTC-PERPETUAL history — candles, raw trades with the liquidation flag, funding — into a local store. Downloads once; a re-run resumes rather than re-fetching.
<code>replay</code>	Runs the real scoring pipeline over that history and produces the verdicts the engine <i>would</i> have called.
<code>validate</code>	The honesty check on <code>replay</code> : joins those synthetic verdicts to real logged ones over the same hours and reports how often they agree.
<code>report</code>	Runs the same Analysis Report you get from the status bar, but over any CSV — this is how you report on a pooled book (your box plus the collector, concatenated).
<code>coverage</code>	Audits the raw-trade store for holes: which UTC hours are properly captured, which are defective, which the collector simply wasn't running for.

How to use it, in order. `fetch` → `replay` → `validate`. Don't skip `validate`. A replay you haven't agreement-checked is a story, not evidence — and the agreement number is the one that tells you whether to believe the rest. Low agreement means distrust the replay, not the engine.

coverage is the one to run before any study that leans on the tape. A missing hour is silent everywhere else — the store just has fewer rows, and nothing complains. Run it, and read the DEFECT count first: those are hours the collector was supposed to have and doesn't. `not-capturing` and `out-of-scope-weekend` hours are expected and fine. Add `--verify-venue` to diff the last 24 hours against Deribit itself; that's the only part that goes to the network.

One trap worth knowing. `coverage` exits 0 by default even when it finds defects — that's deliberate, so running it by hand never looks like a failure. It only fails the run if you pass `--strict`. So read the DEFECT line; don't read the exit code.

What it is not. Same rule as What-If Replay: a promising backtest is evidence for a proposal, never a change in itself. And outcomes here are scored the same way the Analysis Report scores them — a wick **touching** the placed target or stop on a historical bar, with no fills, no slippage and no queue position. Real execution is worse than any number it prints.

Quick Reference — Verdict Action Rules

Verdict (displayed)	Stored / wire	Confidence	Action
STRONG LONG / STRONG SHORT	(same)	HIGH	Primary entry signal
MEDIUM LONG / MEDIUM SHORT	LONG / SHORT	MEDIUM	Valid entry signal
WEAK LONG / WEAK SHORT	(same)	LOW	Informational only — no trade
NO TRADE	(same)	N/A	Stand aside
NO TRADE [WEAK X]	(same)	N/A	Gate fired — stand aside, note the lean

(v55, 2026-07-21: the on-screen MEDIUM x is a display rendering only — the CSV, bridge payload, eval cache, and every string-matching site still carry bare LONG / SHORT for the middle band. Actionability keys **direction + confidence** on the frozen bridge contract are unchanged.)

Quick Reference — Context Tags

Context Tag	What to Do
(no tag)	Setup is confirmed — proceed per verdict
MOMENTUM_FADING	Tighten target, don't add size
FLOW_UNCONFIRMED	Wait for flow confirmation before entering
STRUCTURALLY_WEAK	Skip the trade
BELOW_MIN_MOVE	Stand aside — realistic move is too small to trade

Quick Reference — Regime and Sizing

Regime	Sizing Guidance
TRENDING_UP / TRENDING_DOWN	Full size; only trade in regime direction
RANGE_BOUND	Standard size; both directions valid
TRANSITIONAL	Reduce size; scoring is penalised, conviction is lower